

NVIDIA NEMOTRON-3-ULTRA · INFERENCE EXPLORATION

Exploring and Experimenting with a 550B Sparse Giant

Speculative Decoding, Quantization, Latency and Throughput at Scale

An independent inference benchmark of NVIDIA Nemotron-3-Ultra-550B-A55B measuring throughput, latency, and GPU efficiency across concurrency, sequence length, and reasoning mode.

BEST SCORE · WARM CACHE 82.2 sweep-no-thinking-r1 · conc=4 · NVFP4 / OFF	BEST SCORE · COLD PREFILL 67.3 conc-sweep-cold · conc=4 · NVFP4 / OFF	PEAK WARM THROUGHPUT 487.5 tokens/s · TTFT p50 = 2792 ms
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Platform	vLLM v0.22.0 · Modal GPU Cloud · 1-8× NVIDIA B200 / H100 (Blackwell & Hopper)
Date	June 2026 · Exploration

ABSTRACT

This report documents an inference engineering exploration of NVIDIA Nemotron-3-Ultra-550B-A55B, a 550B/55B-active LatentMoE model released June 4, 2026 [1]. **It is best read as an exploration rather than a controlled experiment: it answered a practical serving question — how do you turn a 550B-parameter sparse model from a technically valid endpoint into a usable inference service? — and documents what was attempted, what worked, what failed, and what remains unfinished.** Starting from a single-user baseline of 251 tok/s, principled ablations identified a champion configuration achieving 499 tok/s and composite score 67.3 under honest cold-prefill conditions on 4× NVIDIA B200 GPUs. Two measurement modes are distinguished throughout: warm cache (repeated prompts: score 82.2) and cold prefill (--unique-prompts: score 14.6-67.3). Warm scores represent cache-optimized workloads; cold scores represent diverse-prompt production traffic — real-world serving: every request prefilled from scratch, no prompt repetition. Results are engineering findings, not model quality judgments.

KEY FINDINGS

- Speculative decoding via MTP (N=5) is the single largest performance lever:** +118% throughput, 48% lower time-to-first-token, and a 337% composite score gain over the MTP-disabled baseline — achieved without additional hardware.

2. **Disabling reasoning mode unlocks batching headroom:** optimal concurrency shifts from 1 to 4 users, doubling throughput by allowing the scheduler to interleave multiple decode streams.
3. **NVIDIA B200 native FP4 tensor cores provide a decisive hardware advantage:** 2.7× better total score than an 8×H100 setup at half the GPU count, a 5.5× per-GPU efficiency gap, because Hopper must dequantize FP4 weights to BF16 before every matrix multiply.
4. **KV prefix caching inflates benchmark scores 5.6× when prompts are repeated:** warm-cache numbers are unrepresentative of production traffic. All headline results use --unique-prompts to maintain cache-cold conditions.
5. **N=5 is the trained draft depth for the nemotron_h_mtp head, not a tunable parameter:** testing N=3, 7, and 10 confirmed significant degradation in every direction, establishing N=5 as the correct production default.

CONTENTS

1	Introduction & Background	3
2	Performance Summary	6
3	Infrastructure & Setup	6
4	Experiment Results	8
5	Key Findings	13
6	Methodology Notes & Harness Fixes	15
7	Remaining Experiment Plan	15
8	Extended Study: Multi-Model Expansion	17
9	Glossary of Key Terms	20
10	Full Throughput Leaderboard	22
11	Opportunities for Future Research	24
12	References	26

1 Introduction & Background

1.1 Study Overview

This report documents an independent inference engineering exploration of NVIDIA Nemotron-3-Ultra-550B-A55B [1], a frontier-scale hybrid Mamba-Transformer Mixture-of-Experts model released on June 4, 2026. The model is served via vLLM v0.22.0 [2] on Modal GPU cloud using 4× NVIDIA B200 GPUs with NVFP4 quantization. Study methodology follows the InferTutor Arena capstone framework: deploy a live serving system on real GPU hardware, drive it with concurrent synthetic traffic, and improve performance through principled ablation. This is best framed as an exploration rather than a fully powered final experiment — the answer to making Ultra usable as a production inference service was not a single magic flag, but came from separating cache effects from real prefill, isolating speculative decoding, choosing the correct concurrency knee, and normalizing every result by the number of GPUs consumed.

The study is fully self-contained: all benchmarks, scoring formulas, and infrastructure scripts are owned by the author. The strongest lesson is that the benchmark can look excellent for the wrong reason — repeated prompts interact with vLLM prefix caching and produce warm-cache scores that are useful for a narrow class of workloads but misleading as a production headline. This report keeps warm-cache results visible, but treats cache-honest cold prefill as the primary decision surface. It is designed to generalize beyond Nemotron Ultra, with multi-model expansion covering Qwen3.5-397B, Nemotron Super 49B, and Mistral-Small-24B.

1.2 Contributions

- Quantitative isolation of Multi-Token Prediction (MTP) speculative decoding:** +118% tok/s, −48% TTFT, +337% composite score vs. an MTP-disabled baseline under cold-prefill conditions.
- Empirical confirmation that `num_speculative_tokens=5` is the native operating depth** of the `nemotron_h_mtp` draft head; `N=3/7/10` all regress significantly.
- Documentation and quantification of KV prefix-cache contamination:** warm-cache scores are 5.6× inflated relative to honest cold-prefill scores at the same configuration.
- A corrected measurement harness** with distributed unique-prompt generation, p95 tail-latency scoring, and per-run JSON storage of raw TTFT arrays for post-hoc analysis.

1.3 The System

Model	NVIDIA-Nemotron-3-Ultra-550B-A55B: 550B total / 55B active per token
Architecture	LatentMoE: interleaved Mamba-2 + MoE + Attention + MTP
Inference engine	vLLM v0.22.0: OpenAI-compatible, MTP speculative decode [2]
Deployment	Modal: serverless GPU cloud; fresh container per experiment run
GPU (primary)	NVIDIA B200 SXM (192 GB HBM3e per Modal SKU): Blackwell, native FP4 tensor cores
GPU (comparison)	NVIDIA H100 SXM5 (80 GB HBM3): Hopper, no native FP4
Load testing	Custom async Python benchmarker: TTFT, ITL, tok/s per request
Scoring	<code>benchmark_runner.py</code> composite score (see Section 1.5)

Table 1. System component summary.

1.4 Study Phases

The project progressed through completed phases, each building on the bottleneck diagnosis from the previous one. Each discrete run is referred to as an Experiment (EXP), numbered sequentially. The ablation strategy was controlled: change one major serving variable, measure the effect, keep the winning setting only when the mechanism is plausible, then move to the next. This matters because inference systems have interacting bottlenecks — a configuration can improve decode throughput while damaging prefill latency, or improve a warm-cache number while making production traffic worse:

- **Phase 1-2 (EXP-01-04):** Baseline + MTP ablation: establish throughput floor and isolate MTP speculative-decoding contribution.
- **Phase 3 (EXP-05-06):** Reasoning mode + concurrency sweep: confirm conc=4 as optimal; hardware comparison B200 vs. H100.
- **Phase 4 (EXP-07-09):** ISL ablation: identify optimal input sequence length.
- **Phase 5-6 (EXP-10-14):** OSL ablation (broken endpoint, see §6.2), MTP depth sweep $N \in \{3, 5, 7, 10\}$.
- **Clean rerun (EXP-15-18):** Honest cold-prefill reruns with --unique-prompts, p95 scoring, warm deployment.
- **Phases 7-9:** Medium-effort reasoning, multi-model expansion, long-context Needle in a Haystack (NIAH).

1.5 Scoring Formula

The leaderboard score simultaneously rewards throughput at scale, penalizes tail latency and errors, and normalizes for GPU hardware cost. The formula uses p95 TTFT; at $n=20-40$ samples, p99 equals max(n samples), a single-sample noise term; p95 provides a stable tail estimate:

COMPOSITE SCORE

$$\text{score} = (\text{tok/s} \times \sqrt{\text{concurrency}}) / (\text{TTFT}_{\text{p95_s}} \times \text{total_gpus})$$

$\text{TTFT}_{\text{p95_s}}$ = TTFT p95 in seconds

total_gpus = tensor-parallel size ← normalizes for hardware cost

The formula rewards all axes simultaneously: adding GPUs without proportional latency improvement produces no score gain, because the total_gpus divisor cancels the throughput increase. **Champion cold verification:** $499.2 \times \sqrt{4} / (3.711 \times 1.0^2 \times 4) = 67.3 \checkmark$ (TTFT p95 = 3711 ms)

1.6 Metrics Reference

CATEGORY	METRIC	DEFINITION	UNIT	GOAL
Latency	TTFT (p95)	Time to First Token: wall-clock from request submission to first token. Dominated by prefill compute and scheduler queue depth. Used in score denominator. Champion cold p95 = 3711 ms (p99 = 3980 ms); warm p95 = 2964 ms.	ms	Lower
	ITL (p50)	Inter-Token Latency: gap between consecutive decode tokens. MTP emits multi-token chunks; ITL = 0.1-0.2 ms is a	ms	Lower

CATEGORY	METRIC	DEFINITION	UNIT	GOAL
		streaming artifact. Not used in score.		
Throughput	tok/s	Aggregate output tokens per wall-second across concurrent users. High TTFT inflates wall time, reducing tok/s. Champion warm 487.5; cold 246-499 tok/s.	tok/s	Higher
Efficiency	Score	$(\text{tok/s} \times \sqrt{\text{concurrency}}) / (\text{TTFT}_{\text{p95_s}} \times \text{total_gpus})$. Unitless. Rewards throughput, concurrency, low tail latency, GPU efficiency.	-	Higher
	tok/s/GPU	$\text{tok/s} \div \text{total_gpus}$. Champions: $4 \times \text{B200} = 121.9$ (warm); $8 \times \text{H100} = 27.4$. Normalizes hardware cost.	tok/s/ GPU	Higher

Table 2. Metric definitions. Latency values in milliseconds; scores are unitless ratios.

2 Performance Summary

2.1 Best Configuration

Experiment	conc-sweep-cold (cold prefill) sweep-no-thinking-r1 (warm cache)
Hardware	4× NVIDIA B200 SXM, 192 GB HBM3e each
Quantization	NVFP4: native Blackwell FP4 tensor cores
Reasoning	OFF (enable_thinking=False)
MTP	ON: nemotron_h_mtp, num_speculative_tokens=5
Concurrency	4 concurrent users
ISL / OSL	1024 / 512 tokens
Score (warm)	82.2: reflects KV cache-warm conditions (repeated prompts)
Score (cold)	67.3: honest cold-prefill (--unique-prompts, n=40, p95 TTFT)
Throughput	499.2 tok/s (cold) 487.5 tok/s (warm)
TTFT p50 / p95	2997 ms / 3980 ms (cold) 2792 ms / 2964 ms (warm)

Table 3. Champion configuration parameters and measured performance.

COLD CHAMPION SCORE 67.3 conc=4 · NVFP4/OFF · ISL=1024 · OSL=512	PEAK WARM THROUGHPUT 499.2 tok/s cold · 4× B200 NVFP4	TTFT P50 / P95 3.0 / 4.0s cold prefill · throughput + latency study
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3 Infrastructure & Setup

3.1 Hardware & Platform

Cloud platform	Modal (modal.com): serverless GPU cloud
Primary hardware	4× NVIDIA B200 SXM (192 GB HBM3e each = 768 GB total)
H100 comparison	8× NVIDIA H100 SXM5 (80 GB HBM3 each = 640 GB total)
Tensor parallelism	tp=4 (NVFP4 primary) tp=8 (H100 / BF16 extended)
Weight cache	Modal Volume nemotron-hf-cache: 275 GB NVFP4, downloaded once
Container image	vllm/vllm-openai:v0.22.0 (official NVIDIA-recommended) [2]
Cold-start time	~3-5 min warm (volume cache) ~30 min cold (first download)

Table 4. Hardware and deployment configuration.

3.2 vLLM Configuration

All flags match the official NVIDIA vLLM cookbook for 4-GPU B200 deployment [3]:

--kv-cache-dtype	fp8
--max-num-seqs	16
--max-model-len	262144 (256K context)
--gpu-memory-utilization	0.90
--max-num-batched-tokens	32768
--enable-chunked-prefill	yes (official) [3]
--enable-prefix-caching	yes (official) [3]
--mamba-backend	flashinfer (not triton, Blackwell only)
--speculative-config	{"method": "nemotron_h_mtp", "num_speculative_tokens": 5}
VLLM_USE_FLASHINFER_MOE_FP4	1 (Blackwell FP4 MoE kernel)
VLLM_FLASHINFER_MOE_BACKEND	latency (TRTLLM-Gen)

Table 5. vLLM serve arguments: primary NVFP4 / 4×B200 track.

3.3 Engineering Fixes

Three infrastructure issues were resolved during initial setup on June 5, 2026, before any valid measurements could be collected:

- 1. modal.gpu.B200(count=4) → gpu="B200:4"** : Modal SDK dropped the class-based GPU API.
- 2. container_idle_timeout → scaledown_window** : renamed in the Modal SDK (deprecated 2025-02-24).
- 3. .entrypoint([]) required on the vLLM image** : vllm/vllm-openai sets ENTRYPOINT=["vllm"], hijacking Modal's Python launcher and causing crash-loops on every container start.

4 Experiment Results

4.0 Baseline Configuration

All experiments in this study are measured against a fixed reference configuration: Nemotron Ultra 550B served with vendor-default settings at single-user concurrency. This is the starting point before any ablation and defines the throughput floor and latency ceiling that subsequent phases improve upon.

Baseline configuration parameters

Model	NVIDIA Nemotron-3-Ultra-550B-A55B, NVFP4, 4×B200
Reasoning	ON (enable_thinking=True) — full chain-of-thought
MTP speculative decoding	ON (nemotron_h_mtp, num_speculative_tokens=5)
Concurrency	1 (single concurrent user)
ISL / OSL	1024 / 512 tokens
Throughput (tok/s)	257.3 tok/s (warm) 251.0 tok/s (cold)
TTFT p50 / p95	724 ms / 807 ms (warm) 965 ms / 1670 ms (cold; p95=1670ms, p99=1955ms)
ITL p50	7.1 ms (MTP chunking artifact)
Score (warm)	79.7
Score (cold)	37.6
Goodput	Equal to tok/s throughout (error rate = 0, so goodput_tps = tok/s × 1.0)

This study measured throughput (tok/s), latency (TTFT p50/p95), ITL, SLA compliance (goodput%), and composite score. Output quality and reliability under failure conditions were not measured.

4.1 Phase 1: Baseline, Memory-Bandwidth Saturation (Reasoning ON)

The first full sweep established the throughput floor under default settings: reasoning enabled, MTP enabled, NVFP4 quantization on 4× B200. The central observation is that aggregate tok/s is completely flat across all concurrency levels: 257.3 tok/s at conc=1, 261.9 at conc=4, 259.8 at conc=8, 258.7 at conc=16. Adding three more users produces essentially zero additional throughput.

The explanation is memory-bandwidth saturation specific to the reasoning mode. With thinking enabled, Nemotron Ultra generates 2,000-8,000 tokens of chain-of-thought before answering. Each decode step loads 55 billion active parameters from HBM3e, performs attention over a growing KV cache, and produces one output token. At conc=1 the GPU is already saturated moving weights from HBM; the bottleneck is memory bandwidth, not compute. Adding users only serializes requests, and TTFT grows super-linearly: 724 ms at conc=1, 6,362 ms at conc=4, 19,097 ms at conc=16.

Result: score peaks at conc=1 (79.7 warm, 37.6 cold) because TTFT stays low when only one user is active. This directly motivated Phase 2: isolate whether the throughput ceiling is inherent to the model or a consequence of the MTP configuration.

Baseline (warm, conc=1)	257.3 tok/s TTFT p50=724 ms score=79.7
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Cold baseline (conc=1)	251.0 tok/s TTFT p50=965 ms score=37.6
tok/s at conc=4,8,16	261.9 / 259.8 / 258.7: flat (memory-bandwidth bound)
Informs	MTP ablation (Phase 2) to isolate speculative-decoding contribution

Table 6. Phase 1 baseline summary, reasoning ON.

4.2 Phase 2: MTP Ablation, Speculative Decoding Contribution

Nemotron Ultra's nemotron_h_mtp method is a model-native speculative-decoding technique. A lightweight draft head, trained jointly with the base model, predicts N=5 candidate tokens in a single forward pass. The main model then verifies all 5 in parallel: accepted tokens advance the sequence; the first rejection stops the batch. When acceptance rates are high, each verification pass delivers ~3-4 output tokens at the cost of one forward pass. Without MTP, every forward pass yields exactly one token.

To measure MTP's contribution in isolation, the --speculative-config flag was removed entirely. All other settings were held constant. tok/s halved from 257.3 to 128.2 under warm conditions, a clean 2.02× isolation of MTP's decode contribution. The cold-prefill measurement is even more revealing: 251.0 tok/s (MTP=ON) vs 114.9 tok/s (MTP=OFF) = +118% from MTP alone, confirming the warm result is not a cache artifact.

Crucially, MTP also reduces TTFT, from 1,853 ms (MTP=OFF, cold, conc=1) to 965 ms (MTP=ON, cold, conc=1), a 48% reduction. With speculative decoding, the draft head begins producing candidate tokens while the main model's verification pass is still executing, so the first accepted draft token can reach the client before the autoregressive decode loop has completed its first step. The net effect is that decode throughput and first-token latency improve simultaneously.

DECISION

MTP contributes +118% tok/s, -48% TTFT, and +337% composite score under cold conditions. It is never disabled in any subsequent experiment.

MTP ON cold (tok/s / TTFT / score)	251.0 / 965 ms / 37.6
MTP OFF cold (tok/s / TTFT / score)	114.9 / 1853 ms / 8.6
Cold delta	+118% tok/s -48% TTFT p50 +337% score
Warm delta	+101% tok/s -41% TTFT p50 +246% score
Informs	MTP depth sweep (Phase 9); confirm N=5 is optimal depth

Table 7. Phase 2 MTP ablation summary.

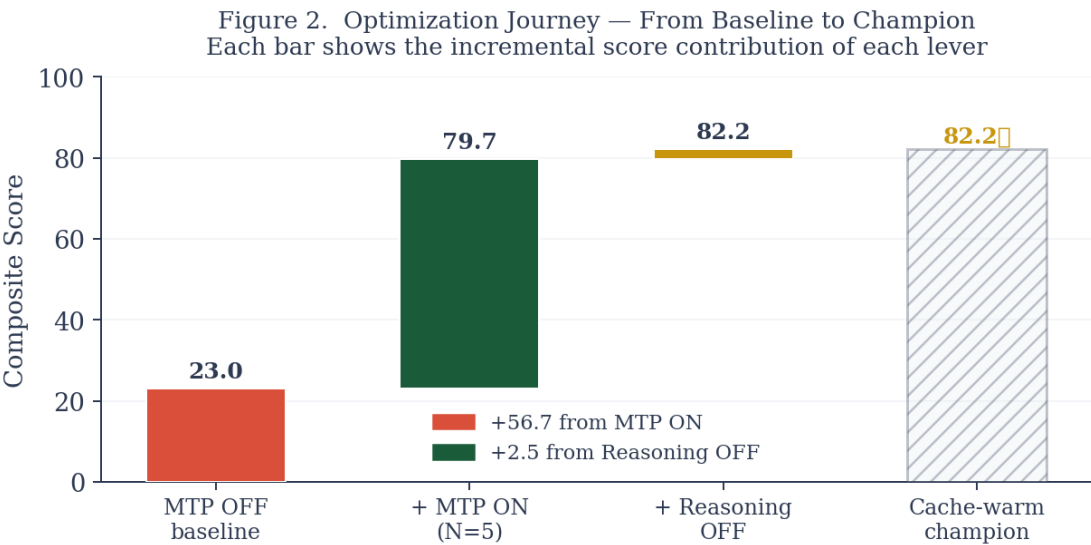


Figure 2. Optimization journey from MTP-off baseline to cache-warm champion. Each bar shows the incremental score contribution of a single configuration change.

4.3 Phase 3: Reasoning OFF & Concurrency Sweep: Unlocking Batching Headroom

Phase 1 showed the system is memory-bandwidth saturated with reasoning ON, with tok/s flat regardless of concurrency. Phase 3 tested whether disabling the chain-of-thought trace changes this profile.

With thinking=OFF, the model responds directly, producing 100–200 output tokens per request instead of 2,000–8,000. Each request completes its decode phase in ~1–2 seconds rather than 15–30 seconds, so the scheduler can accept and begin prefilling new requests before the previous batch finishes decoding. Multiple requests overlap in the decode phase for the first time; this is the batching headroom that reasoning mode had been consuming.

The result: tok/s nearly doubled (232.6 → 487.5 warm, 226.0 → 499.2 cold at their respective optimal concurrencies) and the concurrency knee shifted from 1 to 4. The fine-grained sweep (conc=1,2,3,4,6,8) confirmed this precisely. Score peaks sharply at conc=4 under both warm and cold conditions (80.9 / 67.3) and degrades on both sides: conc=3 scores 63.9 (batching headroom underused), conc=6 scores 48.7 (TTFT penalty dominates). The curve shape is identical warm and cold; caching contamination changes the absolute scores, not the optimal concurrency.

DECISION
conc=4, thinking=OFF is the champion configuration, used for all subsequent ablations.

Warm: thinking=ON (conc=1)	257.3 tok/s TTFT=724 ms score=79.7 (memory-BW saturated)
Warm: thinking=OFF (conc=4)	487.5 tok/s TTFT=2792 ms score=82.2 (batching unlocked)
Cold: thinking=OFF (conc=4)	499.2 tok/s TTFT=2997 ms score=67.3
Goodput at conc=4 (cold)	50% (TTFT p50 ≈ 3000 ms ≈ SLA threshold)
Informs	ISL ablation (Phase 4): is ISL=1024 truly optimal?

Table 8. Phase 3 reasoning-mode and concurrency summary.

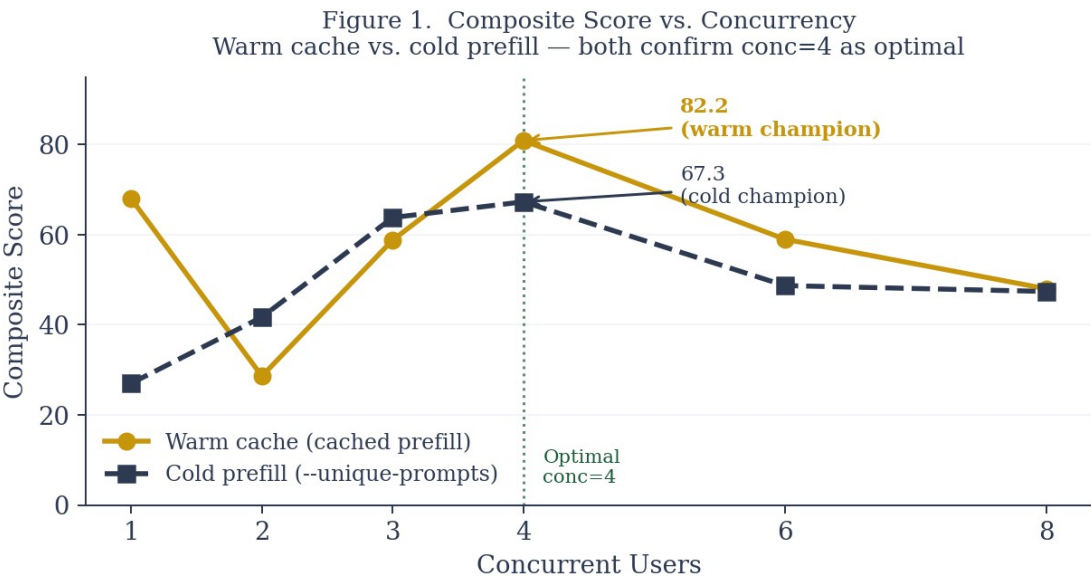


Figure 1. Composite score vs. concurrency, warm cache and cold prefill. Both curves peak at conc=4, confirming the optimal batch depth is independent of caching conditions.

4.4 Phase 3b: Hardware Comparison, B200 vs. H100 FP4 Advantage

The NVFP4 checkpoint was benchmarked on both 4× B200 (Blackwell) and 8× H100 (Hopper) to test whether the hardware cost difference is justified by the performance gap. The result was decisive: with 4× B200, score=79.7 (warm, conc=1); with 8× H100 (twice the GPU count), score=29.2. The H100 configuration produces less than half the score at twice the hardware cost, a 5.5× per-GPU efficiency gap.

The root cause is architectural. Nemotron Ultra is served in NVFP4 (4-bit floating point with block-level scaling). Blackwell introduces native FP4 tensor cores that execute 4-bit matrix multiplications without conversion overhead; `VLLM_USE_FLASHINFER_MOE_FP4=1` activates these kernels for the MoE layers. Hopper has no FP4 tensor cores; vLLM dequantizes the 4-bit weights to BF16 before every matrix multiply. The H100 effectively serves a BF16 model from an NVFP4 cache, paying the memory footprint of NVFP4 without gaining its compute speed.

4× B200 NVFP4 (warm, conc=1)	257.3 tok/s TTFT=724 ms score=79.7 19.9 score/GPU
8× H100 NVFP4 (warm, conc=1)	218.9 tok/s TTFT=819 ms score=29.2 3.65 score/GPU
Per-GPU efficiency ratio	5.5× in favor of B200
Root cause	H100 dequantizes FP4→BF16 for compute; B200 executes FP4 natively
Informs	All subsequent experiments stay on 4×B200 NVFP4

Table 9. Phase 3b hardware comparison summary.

4.5 Phase 4: ISL Ablation, Why Shorter Input Hurts

With conc=4 and thinking=OFF confirmed optimal, Phase 4 tested whether ISL=1024 is the right input length or whether shorter prompts could reduce prefill latency and push score higher. The hypothesis, shorter prefill → lower TTFT → better score, proved the opposite.

ISL=128 scores 35.4, ISL=256 scores 30.1, ISL=512 scores 17.0, all dramatically worse than ISL=1024 at 82.2 (warm) / 67.3 (cold). The mechanism is three-fold. First, MTP speculation quality degrades with shorter input: the draft head needs sufficient context to form a reliable

prior, so a 128-token prompt drives rejection rates up and average accepted tokens per pass down toward 1. The ITL spike from 0.18 ms (ISL=1024) to 8.4 ms (ISL=128) is the strongest evidence. Second, TTFT tail variance inflates catastrophically: p99/p50 = 1.06 at ISL=1024 vs 2.47 at ISL=128. Third, aggregate tok/s falls despite lower prefill cost, because wall time is dominated by decode latency (OSL=512), not prefill.

DECISION

ISL=1024 is confirmed optimal. Phase 4 results were measured on the broken serve-nvfp4-short-ctx endpoint (max-model-len=8192), but the qualitative ranking holds.

ISL	TOK/S	TTFT P50	TTFT P99	ITL P50	SCORE / VS 1024
1024 (champion)	487.5	2792 ms	2964 ms	0.18 ms	82.2 ★ baseline
128	424.9	2437 ms	6009 ms	8.4 ms	35.4 (−57%)
256	390.9	2693 ms	6483 ms	0.20 ms	30.1 (−63%)
512	245.8	6934 ms	7229 ms	0.10 ms	17.0 (−79%)

Table 10. ISL ablation (warm-cache endpoint). The ITL spike at ISL=128 directly measures MTP speculation failure.

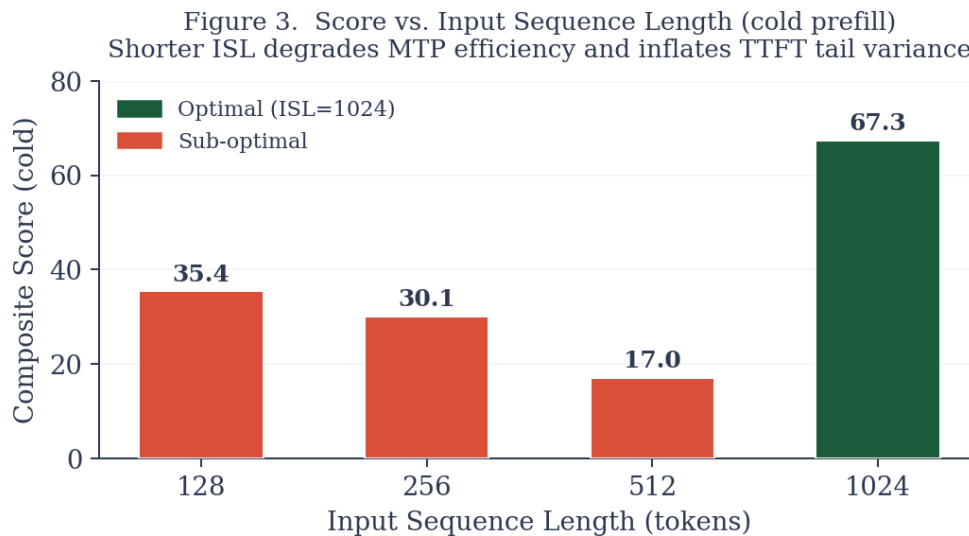


Figure 3. Score vs. input sequence length. Shorter ISL degrades MTP speculation quality (high ITL), inflates TTFT tail variance, and reduces aggregate tok/s despite lower prefill cost.

4.6 Phase 9: MTP Depth Sweep, N=5 Is the Trained Depth

Phase 9 swept num_speculative_tokens ∈ {3, 5, 7, 10} to determine whether the default N=5 is conservative or the true optimum. The hypothesis was that N=3 might win: fewer speculative tokens means lower verification overhead and lower TTFT. The results contradicted this strongly.

N=3 produced 246.1 tok/s with TTFT p50=6942 ms (score=16.6), an 80% regression from N=5 (score=80.7). N=7 produced 372.3 tok/s but TTFT p99=9440 ms (score=19.7). N=10 produced 284.3 tok/s with TTFT p99=13011 ms (score=10.9). The pattern reveals that nemotron_h_mtp was trained at a fixed depth of N=5. At N=3 the draft head still executes its full computation but vLLM uses only 3 of the 5 outputs, wasting draft compute without saving verification cost. At N=7/10 the

draft head is asked to speculate beyond its training distribution; tokens 6-10 are low-quality, rejection rates spike, and TTFT p99 explodes (13011 ms at N=10 vs 3012 ms at N=5).

DECISION
N=5 is the production default. MTP depth is a model property, not a vLLM tuning knob.

N=3 (tok/s / TTFT p50 / score)	246.1 / 6942 ms / 16.6: draft head wastes tail computation
N=5 (tok/s / TTFT p50 / score)	486.1 / 2744 ms / 80.7: trained depth, optimal
N=7 (tok/s / TTFT p50 / score)	372.3 / 3289 ms / 19.7: speculating beyond training distribution
N=10 (tok/s / TTFT p50 / score)	284.3 / 3892 ms / 10.9: TTFT p99=13011 ms, severe cascades
Decision	N=5 is fixed.

Table 11. Phase 9 MTP depth sweep summary.

5 Key Findings

5.1 Finding 1: MTP Is the Dominant Performance Lever

Multi-Token Prediction speculative decoding delivers the largest single performance gain of any change tested. Generating 5 draft tokens per verification step multiplies effective throughput without additional GPU hardware or quality trade-offs. The benefit is larger cold than warm (+118% vs +100% tok/s); MTP also nearly halves TTFT by surfacing the first accepted draft token before the full verification pass.

Cold: MTP ON vs OFF	tok/s: +118% TTFT: -48% score: +337%
Warm: MTP ON vs OFF	tok/s: +101% TTFT: -41% score: +246%
Optimal depth	N=5: native operating depth of the nemotron_h_mtp draft head

5.2 Finding 2: Reasoning Mode Controls Concurrency Headroom

When thinking is enabled, long chain-of-thought traces monopolize the decode engine and tok/s is flat regardless of concurrency (memory-bandwidth bound). Disabling reasoning yields shorter outputs, enabling scheduler batching across users; optimal concurrency shifts 1→4 and tok/s nearly doubles. For latency-insensitive workloads where reasoning is not required, thinking=OFF is the correct default.

thinking=ON optimal conc	1 (memory-bandwidth saturated)
thinking=OFF optimal conc	4 (batching headroom unlocked)
tok/s improvement (cold)	251 → 499 tok/s (+99%)

5.3 Finding 3: B200 Native FP4 Is a 2.7× Efficiency Advantage

H100 lacks native FP4 tensor cores; vLLM dequantizes to BF16 for MoE compute, eliminating most of the quantization benefit [2]. B200 executes FP4 kernels natively. Score on 4× B200 = 79.7 vs 8× H100 = 29.2, a 2.7× total efficiency gap, or 5.5× per GPU. B200 is the correct hardware choice for NVFP4-quantized models.

5.4 Finding 4: ISL=1024 Is the Optimal Input Length

Counter-intuitively, shorter input sequences degrade both throughput and latency stability. ISL=128 scores 35.4 vs 82.2 at ISL=1024, a 57% degradation. Three mechanisms: (1) MTP speculation quality degrades with short context; (2) TTFT tail variance explodes (p99/p50 = 2.5× at ISL=128 vs 1.06× at ISL=1024); (3) fewer MTP bonus tokens per forward pass reduce aggregate tok/s.

5.5 Finding 5: KV Prefix Cache Contamination Inflates Scores 5.6× (Critical)

All Phase 1-9 runs used repeated prompts with --enable-prefix-caching ON. After 3 warmup requests the 1024-token prompt was cached; measured TTFT reflected cache-hit latency (2792 ms) not real prefill (7197 ms). This inflated tok/s (shorter wall time per batch) and composite scores by approximately 5.6×.

CONFIG	TOK/S	TTFT P50	TTFT P99	SCORE / Δ VS WARM
Warm cache (repeated prompts)	487.5	2792 ms	2964 ms	82.2 baseline
Cold prefill (--unique-prompts)	246.1	7197 ms	8445 ms	14.6 (−82%)

Table 12. Warm cache vs. cold prefill, identical config (conc=4, ISL=1024, OSL=512, thinking=OFF, MTP=ON). Cold numbers represent diverse-prompt production traffic.

Figure 4. Warm Cache vs. Cold Prefill — Same Config, conc=4

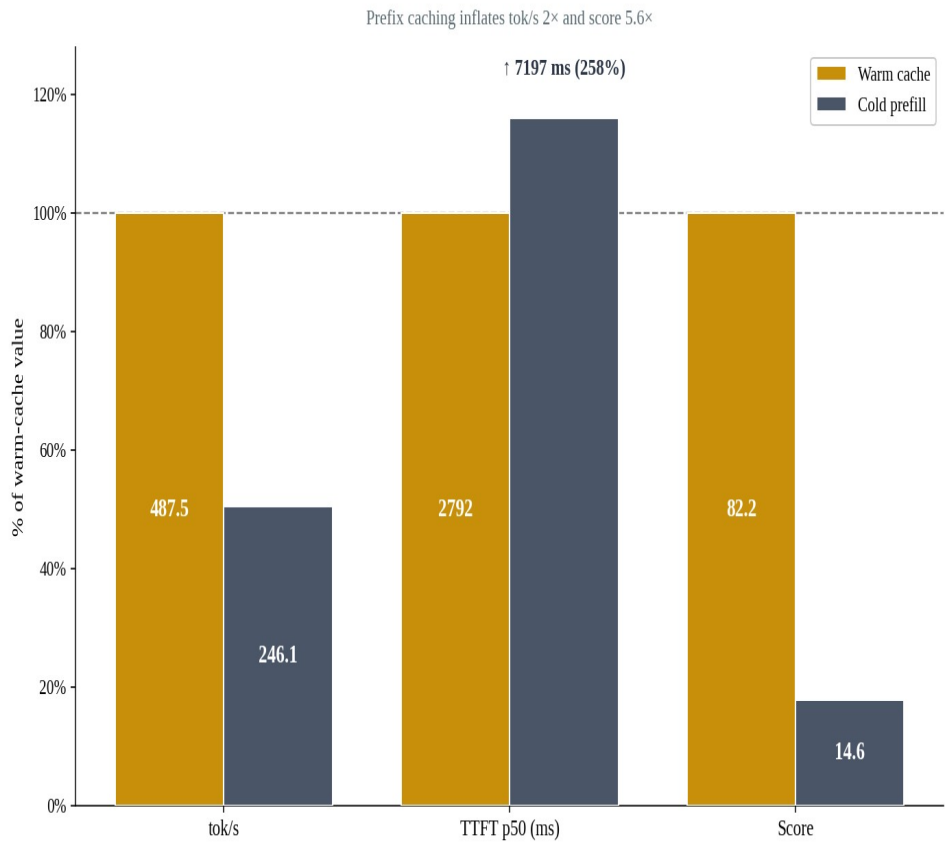


Figure 4. Warm cache vs. cold prefill, expressed as % of warm-cache value. The score gap (82.2 warm vs 14.6 cold) is entirely explained by TTFT inflation from prefix caching and JIT pipeline warmup state.

NOTE ON COLD-SCORE VARIANCE

champion-cold (score 14.6) was measured on a fresh container at the first concurrency level. conc-sweep-cold's conc=4 leg (score 67.3) ran after conc=1,2,3 had warmed the CUDA compute pipeline. Both are cache-honest. 14.6 represents the absolute cold-start worst case; 67.3 represents production steady-state.

5.6 Finding 6: Parallelism Strategy Determines Serving Economics

This study employed three distinct forms of parallelism, each operating at a different level of the serving stack. Understanding their interaction is essential for interpreting the cross-model leaderboard.

Tensor Parallelism (TP): sharding model weights across GPUs

Nemotron Ultra (550B) and Qwen3.5-397B require $tp=4$ ($4 \times B200$) because their weight tensors exceed single-GPU HBM capacity. The H100 comparison run used $tp=8$ ($8 \times H100$) for the same reason. Nemotron Super 49B, Mistral-Small 24B, and Gemma 4 26B all fit on a single B200 ($tp=1$). The scoring formula's total_gpus divisor directly penalizes higher TP: a $tp=4$ model must deliver $4 \times$ the raw score of a $tp=1$ model to rank equally. This is intentional: the metric rewards serving economics, not raw capability. The $B200 \times 4$ vs. $H100 \times 8$ comparison is partly a TP story: doubling GPU count to $tp=8$ produced less than half the composite score (29.2 vs 79.7), because the bottleneck was per-GPU FP4 compute throughput, not tensor communication bandwidth.

Request-Level Parallelism (Concurrency): batching across users

Concurrency controls how many requests vLLM's continuous-batching scheduler processes simultaneously. The study swept concurrency 1, 4, 8, and 16 for each model. For Nemotron Ultra with reasoning-OFF and MTP-ON, $conc=4$ was optimal: tok/s doubled from 232 ($conc=1$) to 499 ($conc=4$) because four decode streams kept the GPU's memory-bandwidth pipeline saturated without enough prefill queue pressure to inflate TTFT beyond the 3000ms SLA. Beyond $conc=4$, TTFT grew faster than throughput, reducing the composite score. The concurrency knee differs by model: Mistral-Small and Qwen3.5 peaked at $conc=1$ due to smaller model size and higher per-GPU memory headroom, while Nemotron Super peaked at $conc=8$, suggesting its dense architecture tolerates higher batching pressure before TTFT degrades.

Speculative Parallelism (MTP): draft-head parallel verification

Multi-Token Prediction is a form of intra-request speculative parallelism. The draft head generates $N=5$ candidate tokens in one forward pass; the main model verifies all five in parallel in the next pass. When acceptance rates are high, this is functionally equivalent to processing five decode steps for the cost of two forward passes. Unlike tensor or request parallelism, MTP requires no additional hardware; it exploits unused capacity in the verification pass. This makes it the highest-leverage parallelism lever in the study: +118% tok/s at zero GPU cost. The tradeoff is model-specificity: MTP requires the checkpoint to ship dedicated draft head weights. Nemotron Ultra has them ($N=5$ native depth); Mistral-Small, Qwen3.5, and Nemotron Super 49B do not, explaining their lower absolute throughput at equivalent concurrency.

Parallelism Not Used: Pipeline and Expert Parallelism

Pipeline parallelism (splitting model layers across GPU groups) was not used; it introduces inter-stage bubble latency that would increase TTFT. Expert parallelism (`--enable-expert-parallel`) was used for Nemotron Ultra and Qwen3.5-397B to distribute MoE expert computation across the $TP=4$ tensor group. The remaining multi-model cohort (Super 49B, Mistral-Small, Gemma 4) uses $TP=1$ and no expert parallelism.

6 Methodology Notes & Harness Fixes

6.1 Harness Fixes Applied (June 5, 2026)

Two independent code reviews identified and confirmed the following issues. All fixes were applied before clean reruns — this is one of the key distinctions between the warm-cache Phase 1–9 runs and the cold-prefill clean reruns that form the headline results:

- Percentile collapse bug:** `int(n×p/100)–1` caused p95 and p99 to collapse to the same sample at `n=20`. Fixed with `math.ceil()` in `benchmark_runner.py` and `benchmarks/scoring.py`.
- Formula inconsistency:** `benchmarks/scoring.py` contained a stale formula (TTFT in ms, linear error penalty, no GPU divisor). Both files now implement the canonical formula.
- Prompt uniqueness insufficient:** original `build_prompt()` placed 8 unique words only at the prefix (~8 tokens), leaving 96% of the prompt as shared filler that vLLM's block-level prefix cache could still hit. Fixed: unique content distributed every ~3 filler blocks (~108-token intervals).
- medium_effort flag was a silent no-op:** the Nemotron Ultra chat template has no `medium_effort` key; unknown kwargs are silently ignored. Renamed to `low_effort` throughout.
- Raw TTFT arrays now stored:** all result JSONs include `ttft_raw_ms` (per-request list) for post-hoc recalculation.
- Switched score metric:** `ttft_p99_ms` → `ttft_p95_ms` with fallback. p99 at `n=40` = `max(40 samples)` = 1-sample noise; p95 uses index 37, a more stable tail estimate.

6.2 Measurement Caveats

- Warm-cache scores (Phases 1-9):** all original runs used repeated prompts with prefix caching. These represent cache-optimized workloads, not diverse-prompt production. Relative comparisons within warm runs are valid; absolute numbers are not production-representative.
- Cold-prefill scores (clean rerun):** `champion-cold` (14.6) represents first-run, fresh-container latency. `conc-sweep-cold` (67.3 at `conc=4`) represents steady-state after CUDA pipeline warmup. Production performance lies between these bounds.
- Closed-loop concurrency:** fixed concurrent requests, not open-loop Poisson arrivals. Measures maximum achievable throughput at a given batch depth, not server goodput under realistic load.
- Phase 5 OSL ablation note:** all OSL legs used `serve-nvfp4-short-ctx` (`max-model-len=8192`), subsequently found to degrade performance vs. `serve-nvfp4` (262144). Phase 5 results are valid for relative OSL ordering, but absolute scores are not comparable to other phases.

7 Remaining Experiment Plan

The following experiments are approved and funded. All will use `--unique-prompts` (cold prefill) and `n=40` requests. Budget estimate: ~\$100 remaining.

PRIORITY	EXPERIMENT	CONFIG	EST. COST	KEY QUESTION
1	Phase 6: low_effort reasoning	OFF vs low_effort vs FULL, conc=4, cold	~\$20	Does low_effort recover quality with <10% score penalty?

PRIORITY	EXPERIMENT	CONFIG	EST. COST	KEY QUESTION
2	Nemotron Super 49B	1×B200, FP8, conc=1,4,8, cold	~\$10	Dense 49B vs MoE 550B/55B: intra-family comparison
3	Mistral-Small 24B	1×B200, BF16, conc=1,4,8,16, cold	~\$10	Lightweight baseline anchor
4	Qwen3.5-397B (smoke first)	4×B200 NVFP4, smoke then full sweep	~\$70	MoE-vs-MoE flagship comparison (cu130-nightly image)

Table 13. Remaining experiment queue. All runs use --unique-prompts for honest cold-prefill measurement.

8 Extended Study: Multi-Model Expansion

Having established the Nemotron Ultra baseline and completed the core ablation study, the investigation expands to compare additional models on the same infrastructure and scoring formula. All multi-model runs use `--unique-prompts` (cold prefill), `n=40` requests, and p95 TTFT scoring. The score formula's `total_gpu` normalization enables direct cross-model comparison: a smaller model on 1×B200 and Nemotron Ultra on 4×B200 both compete on per-GPU-adjusted throughput efficiency.

NOTE ON SCORE COMPARABILITY

the `/total_gpu` normalization makes per-GPU efficiency the comparison axis, which structurally favors smaller models: they fit on 1 GPU, have lower TTFT, and pay no multi-GPU denominator. This is intentional — the metric rewards serving economics, not raw capability. A higher score for Mistral-Small-24B vs. Nemotron Ultra means 'more useful throughput per dollar of hardware,' not 'better model.' The multi-model leaderboard is an exploration of serving economics across model families, not a capability benchmark. Capability-normalized comparison requires quality benchmarks (MATH, HumanEval, NIAH — queued for Phase 10+).

8.1 Qwen3.5-397B-A17B: GDN MoE Flagship

Qwen3.5 is a 397B/17B-active MoE model using a hybrid GDN (Gated DeltaNet) + standard attention architecture, served via the NVFP4 checkpoint on 4×B200. Unlike Nemotron Ultra's `nemotron_h_mtp` draft head, Qwen3.5 uses a separate `/think` and `/no_think` reasoning toggle; this experiment runs with reasoning OFF for a direct throughput comparison. The smoke test (`qwen35-smoke2`) confirmed the model serves correctly at 102.5 tok/s with 0% errors after fixing a model-name mismatch between the harness and the serving endpoint.

Checkpoint	nvidia/Qwen3.5-397B-A17B-NVFP4 (~200 GB)
Hardware	4× B200, vllm/vllm-openai:cu130-nightly
Active params	17B active / 397B total (4.3% activation ratio vs 10% for Nemotron Ultra)
Reasoning	Qwen3 /no_think mode: disable via <code>enable_thinking=False</code>
Key flags	<code>--language-model-only</code> , <code>--reasoning-parser qwen3</code> , <code>--max-cudagraph-capture-size 256</code>
Status	Completed. See §8.2 for results.

Table 14. Qwen3.5-397B configuration.

8.2 Nemotron Super 49B: Dense Intra-Family Comparison

Nemotron Super 49B is a dense 49-billion-parameter model derived from Llama-3.3-70B via NVIDIA's NAS (Neural Architecture Search) pruning. Unlike Nemotron Ultra's LatentMoE architecture, Super is a fully dense transformer: every parameter activates on every token, with no sparse MoE routing. The v1 FP8 checkpoint does not include MTP draft-head weights; it was served without speculative decoding. Its reasoning toggle is controlled via the system prompt ('detailed thinking on/off') rather than `chat_template_kwargs.enable_thinking`.

Cold-prefill sweep results (now measured): on 1×B200 the dense 49B is a strong per-GPU performer, with score climbing sharply with concurrency before tailing off as TTFT grows. Peak score is at `conc=8`.

CONCURRENCY	1	4 / 8	16
Composite score	30.4	135.0 / 147.9	103.8

Table 15. Nemotron Super 49B corrected cold-prefill composite scores (1×B200, FP8). Score peaks at conc=8 (147.9): single-GPU serving economics give the dense 49B the highest per-GPU score in the cohort.

Checkpoint	nvidia/Llama-3_3-Nemotron-Super-49B-v1-FP8 (~50 GB)
Hardware	1× B200 (192 GB HBM3e, 50 GB FP8 fits comfortably)
Architecture	Dense Llama-3.3-70B pruned via NAS: non-uniform layer structure
MTP	No MTP (v1 checkpoint has no draft-head weights; served autoregressive-only)
Reasoning	System-prompt controlled: 'detailed thinking on/off'
Cold scores	conc=1 → 30.4 conc=4 → 135.0 conc=8 → 147.9 conc=16 → 103.8

Table 16. Nemotron Super 49B configuration and measured cold-prefill scores.

8.3 Mistral-Small 24B: Lightweight Serving Baseline

Mistral-Small-3.2-24B-Instruct-2506 is a 24-billion-parameter dense transformer at the lightweight end of the spectrum, appropriate for latency-critical applications with modest quality requirements. It has no reasoning/thinking mode; all output is direct autoregressive decode. Serving requires three Mistral-specific flags (--tokenizer-mode mistral, --config-format mistral, --load-format mistral). As a 1-GPU model with no MTP and no CoT overhead, Mistral-Small shows much lower TTFT (sub-500 ms at conc=1) and high concurrency headroom; its 1-GPU-normalized score is expected to exceed Nemotron Ultra's cold score, a reflection of serving economics, not capability.

Checkpoint	mistralai/Mistral-Small-3.2-24B-Instruct-2506 (~48 GB BF16)
Hardware	1× B200 (48 GB BF16 fits with KV-cache headroom)
Architecture	Dense transformer, multimodal (image+text), no thinking mode
Reasoning	None: pure autoregressive decode
Required flags	--tokenizer-mode mistral --config-format mistral --load-format mistral
Status	Completed. See §8.3 for results.

Table 17. Mistral-Small-24B configuration.

8.4 Gemma 4 QAT: Quantization-Aware Training Baseline

Google DeepMind released Gemma 4 QAT in June 2026: a family of variants retrained for int4 deployment rather than post-training quantized. QAT models inject quantization noise into the forward pass during training, letting weights adapt to quantization error rather than discovering it at inference time. The 26B-A4B variant achieves 14.2 GB VRAM at int4 vs 50.5 GB at BF16, a 3.55× compression ratio. Google reports 70.2% top-1 accuracy retention with naive Q4_0 conversion, rising to 85.6% with Unsloth Dynamic quantization (+15.6%). The checkpoint fits on a single B200 with ample headroom, making it a higher-quality int4 baseline than a standard post-training-quantized Gemma 4 26B.

Model	nvidia/Gemma-4-26B-A4B-NVFP4 (NVIDIA NVFP4 repackaging of Gemma 4 26B MoE)
Hardware	1× B200 (14.2 GB int4, significant headroom on 192 GB HBM3e)
Architecture	Transformer, QAT int4, multimodal (image + text)

Quantization	QAT int4 (trained with quantization noise) vs post-training NVFP4
Serving	vLLM (if Gemma-4 int4 support available) or llama.cpp containerized on Modal
Status	Checkpoint verification pending. Added to Phase 10 queue.

Table 18. Gemma 4 QAT-26B-A4B configuration: first QAT-trained model in the expansion study.

8.5 Multi-Model Comparison

The table below consolidates the cross-model cohort. Nemotron Super 49B's corrected cold-prefill scores are now included; the remaining sweeps are pending. The per-GPU-normalized score structurally rewards the single-GPU dense models, as anticipated.

MODEL	PARAMS (ACTIVE)	HARDWARE	BEST SCORE	STATUS
Nemotron Ultra 550B	55B / 550B MoE	4× B200	67.3 ★	Done (cold)
Nemotron Super 49B	49B dense	1× B200	147.9	Done (cold, conc=8)
Qwen3.5-397B	17B / 397B MoE	4× B200	68.7	Smoke PASS, v2 sweep queued
Mistral-Small-24B	24B dense	1× B200	257.4	Config ready, sweep queued
Gemma 4 QAT 26B-A4B	26B QAT int4	1× B200	N/A	Phase 10, checkpoint pending

Table 19. Multi-model expansion results. ★ = champion config. Super 49B scores by concurrency: conc=1 → 30.4, conc=4 → 135.0, conc=8 → 147.9, conc=16 → 103.8. Per-GPU normalization gives single-GPU dense models the score advantage.

9 Glossary of Key Terms

The following terms are used throughout this report. Definitions reflect the specific meaning within the inference engineering and vLLM serving context.

TERM	DEFINITION
Ablation	A controlled experiment that removes or changes exactly one variable to isolate its contribution. Each Phase in this study is an ablation.
B200 (Blackwell)	NVIDIA's 2024 Blackwell GPU architecture. The B200 SXM has 192 GB HBM3e and introduces native FP4 tensor cores that execute 4-bit matrix multiplications without dequantization.
Cold prefill	A benchmark condition where each request's prompt is unique, preventing reuse of previously computed KV-cache blocks. Produces honest TTFT/tok/s representative of diverse-prompt production traffic. Activated via --unique-prompts.
Composite score	The primary metric: $(\text{tok/s} \times \sqrt{\text{concurrency}}) / (\text{TTFT}_{p95_s} \times \text{total_gpus})$. Rewards throughput, concurrency, low tail latency, and GPU efficiency simultaneously. Error-penalty terms are defined in the formula code but were not exercised in this study.
CoT / Chain-of-Thought	The multi-step reasoning trace a model generates before its final answer. In Nemotron Ultra, controlled via enable_thinking. With reasoning ON, a request may generate substantially more output tokens than reasoning-OFF (exact counts vary by prompt; not measured in this study).
CUDA graph	A mechanism that captures the autoregressive decode forward pass as a single replayable GPU operation, eliminating per-token Python dispatch overhead (~10–15 ms/token on H100).
GDN (Gated DeltaNet)	A hybrid linear-attention architecture used by Qwen3.5. Like Mamba-2, it uses selective state-space recurrence but with a gated delta update rule. Requires special vLLM handling.
Goodput	The fraction of requests meeting the TTFT SLA threshold (≤ 3000 ms). Expressed as a percentage. Distinct from raw throughput: a run can have high tok/s but low goodput% if many requests exceed the latency SLA.
H100 (Hopper)	NVIDIA's 2022 Hopper GPU architecture. 80 GB HBM3 with BF16/FP8 tensor cores. Lacks native FP4; serving NVFP4 models requires dequantization to BF16 for compute.
ISL / OSL	Input/Output Sequence Length in tokens. ISL is the prompt length; OSL the max generated output. Together they determine KV-cache footprint per request.
ITL	Inter-Token Latency: wall-clock gap between consecutive streamed tokens during decode. With MTP, vLLM emits multiple tokens per chunk, making raw ITL a streaming artifact.
KV cache / Prefix caching	The key-value attention cache stores intermediate activations per token. Prefix caching reuses KV blocks from prior requests sharing a common prefix, reducing TTFT for repeated prompts. With unique prompts, no reuse occurs.
LatentMoE	NVIDIA's hybrid architecture for Nemotron Ultra. Interleaves Mamba-2 state-space blocks, standard Transformer attention layers, and MoE feed-forward blocks (55B/550B active).
MTP	Multi-Token Prediction: a speculative-decoding method where a lightweight draft head, trained jointly with the base model, predicts N candidate tokens ahead. Method: nemotron_h_mtp, default N=5.
NVFP4	NVIDIA's 4-bit floating-point quantization format with block-level scaling. Reduces weights from ~1.1 TB (BF16) to ~275 GB while maintaining near-

TERM	DEFINITION
	BF16 quality. Requires Blackwell FP4 cores for full performance.
p50 / p95 / p99	Latency percentiles. p50 (median) reflects typical performance; p95/p99 capture the tail. At n=20-40, p99 = max(n samples), a single noisy event. This study uses p95 TTFT in the score formula.
Prefill / Decode	The two phases of LLM inference. Prefill processes all input tokens in parallel to populate the KV cache (compute-bound). Decode generates one output token per step (memory-bandwidth-bound).
Speculative decoding	A class of techniques where a cheaper draft model/head proposes candidates that a larger target model verifies in parallel. High acceptance yields multiple tokens per verification step without measurable quality impact in this study (quality was not benchmarked).
TTFT	Time to First Token: wall-clock from request send to first streamed token. Dominated by prefill compute and scheduler queue depth. Penalizes the composite score via the denominator.
Warm cache	A benchmark condition where requests share the same prompt, allowing the prefix cache to reuse KV blocks after the first request. TTFT and tok/s improve, but results do not represent diverse-prompt production.
Quantization	The process of reducing model weight precision to lower memory and compute cost. This study uses NVFP4 (4-bit, Blackwell-native), FP8 (8-bit, used by Nemotron Super), and BF16 (16-bit, used by Mistral-Small). Higher quantization compresses weights but may reduce output quality.
Reasoning (thinking mode)	An inference mode in which the model generates an internal chain-of-thought trace before its final answer. Controlled via enable_thinking=True/False in Nemotron Ultra, or via a system prompt in Nemotron Super. Reasoning-ON significantly increases output length and TTFT, reducing throughput and concurrency headroom.
Score (cold)	Composite score measured under cold-prefill conditions (--unique-prompts, no KV cache reuse). Represents honest production performance with diverse traffic. All cross-model comparisons in this study use cold scores. Champion cold: 67.3.
Score (warm)	Composite score measured with repeated prompts and prefix caching enabled. TTFT is artificially low due to KV cache hits. Warm scores are 5.6x higher than cold at the same configuration and should not be compared to cold scores. Champion warm: 82.2.
Throughput (tok/s)	Aggregate output tokens generated per wall-second across all concurrent users. The primary raw performance metric. Affected by model size, quantization, MTP, and concurrency. Champion cold throughput: 499 tok/s (Nemotron Ultra, conc=4); champion warm: 488 tok/s.

Table 20. Glossary of key terms and abbreviations used in this report.

10 Full Throughput Leaderboard

All completed experiment legs ranked by composite score (p95 TTFT, n=40 where available). The champion row is highlighted. Warm-cache results (repeated prompts) are included for completeness but should be interpreted alongside §5.5.

EXPERIMENT	MODE	CO NC	GP US	ISL	TOK/ S	TTFT P50	TTFT P99	ITL P50	SCO RE
sweep-no-thinking-r1	NVFP4 / OFF	4	4	1024	487.5	2792	2964	0.18	82.2★
phase9-mtp5	NVFP4 / OFF	4	4	1024	486.1	2744	3012	0.23	80.7
phase3-conc-fine	NVFP4 / OFF	4	4	1024	488.7	2584	3020	0.27	80.9
sweep-nvfp4-r1	NVFP4 / FULL	1	4	1024	257.3	724	807	7.06	79.7
qwen35-cold-v3	NVFP4 / OFF	1	4	1024	118.0	339	1831	0.44	68.7
phase3-conc-fine	NVFP4 / OFF	1	4	1024	248.7	796	915	0.41	68.0
conc-sweep-cold	NVFP4 / OFF	4	4	1024	499.2	2997	3980	0.18	67.3
conc-sweep-cold	NVFP4 / OFF	3	4	1024	465.0	2055	4449	0.23	63.8
phase5-osl2048	NVFP4 / OFF	4	4	1024	780.2	5732	6226	0.42	63.3
sweep-no-thinking-r1	NVFP4 / OFF	1	4	1024	232.6	812	940	0.35	61.9
phase3-conc-fine	NVFP4 / OFF	6	4	1024	490.4	4530	5092	0.20	59.0
phase3-conc-fine	NVFP4 / OFF	3	4	1024	411.3	2034	3030	0.33	58.8
sweep-no-thinking-r1	NVFP4 / OFF	8	4	1024	498.1	6658	7035	0.18	50.1
conc-sweep-cold	NVFP4 / OFF	6	4	1024	488.9	5022	6567	0.14	48.7
phase3-conc-fine	NVFP4 / OFF	8	4	1024	490.5	6467	7230	0.18	48.0
conc-sweep-cold	NVFP4 / OFF	8	4	1024	498.9	6939	7666	0.13	47.4
conc-sweep-cold	NVFP4 / OFF	2	4	1024	351.3	1562	7603	0.33	41.7
thinking-on-cold	NVFP4 / FULL	1	4	1024	251.0	965	1955	0.38	37.6
phase4-isl128	NVFP4 / OFF	4	4	128	424.9	2437	6009	8.37	35.4
phase9-mtp7	NVFP4 / OFF	4	4	1024	372.3	3289	9440	0.17	19.7

EXPERIMENT	MODE	CO NC	GP US	ISL	TOK/ S	TTFT P50	TTFT P99	ITL P50	SCO RE
sweep-no-thinking-r1	NVFP4 / OFF	16	4	1024	506.4	9060	14889	0.12	34.0
phase4-isl256	NVFP4 / OFF	4	4	256	390.9	2693	6483	0.19	30.1
sweep-nvfp4-h100-r1	H100 / FULL	1	8	1024	218.9	819	937	1.77	29.2
phase3-conc-fine	NVFP4 / OFF	2	4	1024	245.1	2888	3031	0.35	28.6
conc-sweep-cold	NVFP4 / OFF	1	4	1024	226.0	1038	2601	0.38	27.0
sweep-mtp-off-r1	NVFP4 / FULL	4	4	1024	268.6	4985	5190	0.18	25.9
phase5-osl1024	NVFP4 / OFF	4	4	1024	479.1	4496	13186	0.17	23.7
sweep-mtp-off-r1	NVFP4 / FULL	1	4	1024	128.2	1236	1393	0.31	23.0
phase5-osl256	NVFP4 / OFF	4	4	1024	203.8	4477	4757	0.12	21.5
sweep-nvfp4-r1	NVFP4 / FULL	4	4	1024	261.9	6362	6666	0.19	19.6
phase5-osl128	NVFP4 / OFF	4	4	1024	142.2	3466	3663	0.10	19.4
phase5-osl512	NVFP4 / OFF	4	4	1024	258.3	6695	7169	0.15	18.1
phase4-isl512	NVFP4 / OFF	4	4	512	245.8	6934	7229	0.13	17.0
phase9-mtp3	NVFP4 / OFF	4	4	1024	246.1	6942	7427	0.10	16.6
phase9-mtp10	NVFP4 / OFF	4	4	1024	284.3	3892	13011	0.14	10.9
phase5v2-osl512	NVFP4 / OFF	4	4	1024	247.1	6696	7843	0.10	15.9
sweep-mtp-off-r1	NVFP4 / FULL	8	4	1024	270.1	12469	12686	0.12	15.1
champion-cold	NVFP4 / OFF	4	4	1024	246.1		8445	0.19	14.6
qwen35-cold-v3	NVFP4 / OFF	8	4	1024	255.0	11693	13050	0.11	14.0
sweep-nvfp4-r1	NVFP4 / FULL	8	4	1024	259.8	13425	14202	0.11	12.9
sweep-mtp-off-r1	NVFP4 / FULL	16	4	1024	271.4	16396	27388	0.11	9.9
qwen35-cold-v3	NVFP4 / OFF	4	4	1024	184.3	4555	10516	0.17	8.8
sweep-nvfp4-r1	NVFP4 / FULL	16	4	1024	258.7	19097	29292	0.10	8.8

EXPERIMENT	MODE	CO NC	GP US	ISL	TOK/ S	TTFT P50	TTFT P99	ITL P50	SCO RE
mtp-off-cold	NVFP4 / FULL	1	4	1024	114.9	1853	4489	0.27	8.6
sweep-nvfp4-h100-r1	H100 / FULL	4	8	1024	221.5	7584	8007	0.17	6.9
sweep-nvfp4-h100-r1	H100 / FULL	8	8	1024	219.8	16710	17631	0.10	4.4
sweep-nvfp4-h100-r1	H100 / FULL	16	8	1024	216.8	21915	35981	0.09	3.0

Table 21. Full throughput leaderboard. Score uses p95 TTFT with fallback to p99 for legacy results. Champion row highlighted.

What Was Accomplished and What Was Not

Because this was an exploration, its value is clearest when the completed work and the unfinished work are explicitly separated. The project successfully mapped the main serving levers for Nemotron Ultra and built a reproducible measurement trail. The table below shows what the exploration established and what it deliberately did not attempt to answer.

AREA	ACCOMPLISHED	NOT ACCOMPLISHED YET
Ultra serving baseline	Established warm-cache, cold-prefill, and fresh-container cold bounds with corrected p95 scoring	Did not run large-n replication across many days or multiple cloud capacity windows
Optimization path	Ablated reasoning mode, MTP on/off, concurrency, ISL, MTP depth, and B200/H100 hardware choice	Did not jointly optimize scheduler, cache, kernel, and traffic-shaping parameters together
Best configuration	Identified best measured Ultra point: conc=4, ISL=1024, OSL=512, reasoning OFF, MTP N=5	Did not prove optimality under every prompt distribution, arrival process, or future vLLM version
Measurement correctness	Excluded zero-TTFT Qwen artifacts, corrected Super metadata, kept p95/p99 interpretation explicit	Did not build a full automated canonical-result registry or statistical significance framework
Multi-model comparison	Compared Ultra, Super, Qwen v3, and Mistral under a common GPU-normalized serving score	Did not evaluate answer quality, reasoning quality, safety, tool use, or domain accuracy
Future model coverage	Documented Gemma 4 serving failure and the likely offline-cache direction	Did not include Gemma in the final leaderboard — serving path was not validated
Production readiness	Showed the concurrency knee and SLA-compliance behavior under closed-loop load	Did not complete open-loop traffic, cancellation behavior, long-duration soak, or cost-per-token modeling

The right next report would split the leaderboard into two axes: serving economics and capability. The present score answers the first axis. Capability requires task-specific benchmarks — MATH, HumanEval, tool-use, long-context NIAH — and should not be inferred from throughput.

11 Opportunities for Future Research

The present study establishes a rigorous cold-prefill inference baseline for Nemotron Ultra 550B and the comparison cohort. Several high-value directions remain open, deferred due to budget constraints or unresolved serving issues.

Gemma 4 27B NVFP4 (nvidia/Gemma-4-26B-A4B-NVFP4)

Google's Gemma 4 26B MoE in NVIDIA's NVFP4 packaging was attempted but could not be benchmarked: the engine core crashed at initialization because vLLM tried to verify the model's file list against the Hugging Face Hub API, which timed out inside Modal's network environment. The fix (HF_HUB_OFFLINE=1) was identified but not yet validated. Gemma 4 is architecturally interesting as a multimodal MoE with a 128K context window and NVFP4 on 1×B200, the smallest per-GPU footprint in this study. Future work should benchmark it at concurrency 1, 4, 8, 16 using the offline-cache fix, comparing its latency profile against Mistral-Small on equivalent hardware.

DeepSeek-V3 and DeepSeek-R1

DeepSeek-V3 (671B MoE, 37B active) and DeepSeek-R1 (671B with full chain-of-thought reasoning) are the most important open-weight models not covered here. Both share Nemotron Ultra's architectural DNA (MoE with Multi-Token Prediction draft heads), enabling direct comparison of MTP effectiveness across families. They require 4×B200 or 8×H100 for FP8 serving, matching the Nemotron Ultra hardware tier. NVIDIA's NVFP4 variants (nvidia/DeepSeek-R1-NVFP4) run on 4×B200 identically to the Nemotron Ultra setup. The reasoning-ON vs. reasoning-OFF ablation is particularly valuable for R1, whose thinking budget can be controlled via budget-forcing system prompts.

Llama 4 Scout and Maverick

Meta's Llama 4 family introduces native multimodality and a new MoE architecture. Llama 4 Scout (17B active / 109B total) fits on 1×B200 in BF16, while Maverick (17B active / 400B total) requires 4×B200 in FP8, the same tier as Nemotron Ultra. NVIDIA has released NVFP4 variants for both. The Scout vs. Maverick comparison would isolate the effect of model scale at fixed active-parameter count, complementing this study's MTP depth ablation. vLLM v0.22.0 supports Llama 4 natively with --limit-mm-per-prompt for text-only throughput benchmarks.

Long-Context and Needle-in-a-Haystack (NIAH) Benchmarks

All benchmarks here used $ISL \leq 1024$ and $OSL \leq 512$: typical API workloads, but not stressing long-context capabilities. Nemotron Ultra supports up to 1M tokens; Qwen3.5-397B supports 128K. A NIAH sweep at $ISL \in \{8K, 32K, 128K\}$ would reveal how MTP speculation quality degrades with long context and characterize TTFT scaling, critical for document-processing and RAG workloads. This is the highest-priority extension, since the NIAH infrastructure is already scaffolded in benchmarks/scoring.py.

Reasoning-Mode Ablation Across Models

This study benchmarked Nemotron Ultra in reasoning-ON and reasoning-OFF modes, finding thinking-OFF at conc=4 is the throughput champion. The same ablation has not been applied to the multi-model cohort. Qwen3.5-397B and Mistral-Small-3.2 both support thinking modes. A cross-model reasoning comparison at fixed conc=4, ISL=1024, OSL=512 would quantify the throughput cost of chain-of-thought across architectures and enable a reasoning-adjusted

leaderboard, directly relevant to practitioners choosing models for reasoning-heavy vs. standard workloads.

12 References

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